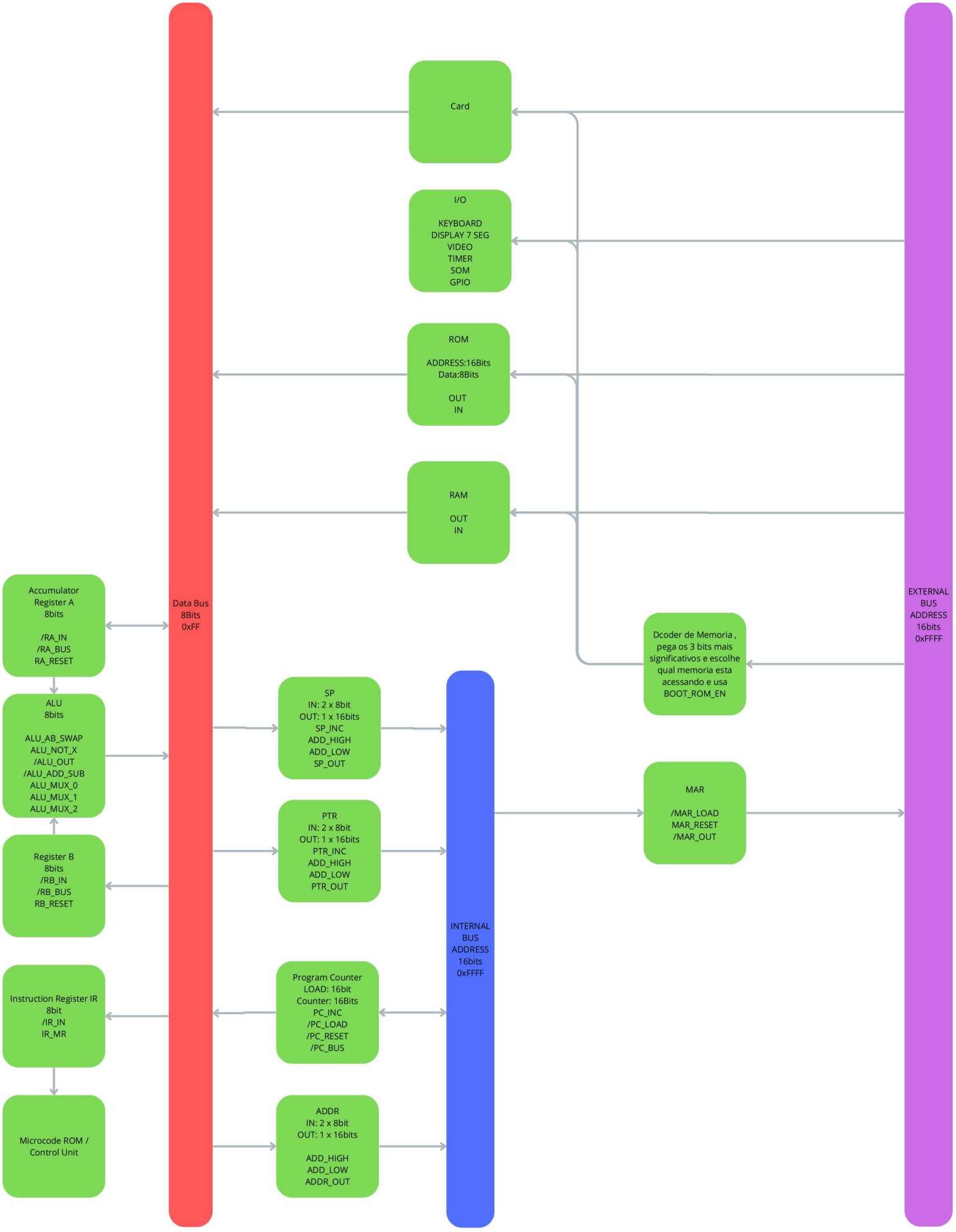

Card
I/O
KEYBOARD
DISPLAY 7 SEG
VIDEO
TIMER
SOM
GPIO
ROM
ADDRESS:16Bits
Data:8Bits
OUT
IN
RAM
OUT
IN
Accumulator
Register A
8bits
/RA_IN
/RA_BUS
RA_RESET
Data Bus
8Bits
0xFF
EXTERNAL
BUS
ADDRESS
16bits
0xFFFF
Dcoder de Memoria ,
pega os 3 bits mais
significativos e escolhe
qual memoria esta
acessando e usa
BOOT_ROM_EN
ALU
8bits
ALU_AB_SWAP
ALU_NOT_X
/ALU_OUT
/ALU_ADD_SUB
ALU_MUX_0
ALU_MUX_1
ALU_MUX_2
SP
IN: 2 x 8bit
OUT: 1 x 16bits
SP_INC
ADD_HIGH
ADD_LOW
SP_OUT
MAR
/MAR_LOAD
MAR_RESET
/MAR_OUT
Register B
8bits
/RB_IN
/RB_BUS
RB_RESET
PTR
IN: 2 x 8bit
OUT: 1 x 16bits
PTR_INC
ADD_HIGH
ADD_LOW
PTR_OUT
INTERNAL
BUS
ADDRESS
16bits
0xFFFF
Instruction Register IR
8bit
/IR_IN
IR_MR
Program Counter
LOAD: 16bit
Counter: 16Bits
PC_INC
/PC_LOAD
/PC_RESET
/PC_BUS
Microcode ROM /
Control Unit
ADDR
IN: 2 x 8bit
OUT: 1 x 16bits
ADD_HIGH
ADD_LOW
ADDR_OUT